

Pseudo-Random number sequence: [62 4 99 60 12 91 21 79 33 71 94 47
10 48 29 72 68 41 73 62 18 88 91 27 42 21 55 51 8 87 19 86
79 14 13 95 7 76 53 41 14 54 32 67 21 87 73 12 38 98]

1. Provide an algorithm to draw samples from $\text{Gamma}(a, 0.5)$ distribution (a is an integer) using Box-Muller transform [Hint: remember chi-square distribution]
2. You want to simulate a distribution $\text{Ga}(10,3)$ using Rejection Sampling. Choose a suitable Gaussian distribution as “wrapper” and demonstrate 5 rounds of sampling. What is the proportion of rejected samples?
3. In the previous case, let us say you are interested to estimate only the expected value instead of the full distribution. How will you do it efficiently through sampling?
4. We want to sample a Beta distribution with parameters (20,20) using Metropolis-Hastings with a Gaussian proposal distribution $N(x,1)$ with initial value 0.5. Estimate proportion of rejections after burn-in phase is over. [sachin..](#)
5. Draw 10 samples $X=(x_1, x_2, \dots, x_{10})$ from $\text{Beta}(5,10)$, using rejection sampling (you can ignore the denominators of Beta PDF for sampling). Use the pseudorandom number sequence above for this purpose. Using each of the above sampled values, draw samples $Y=(y_1, y_2, \dots, y_{10})$ such that $y_i \sim N(x_i, 4)$ using the same pseudorandom number sequence.
6. Consider the following function: $f(x) = 0.3 \cdot N(x, -5, 25) + 0.4 \cdot \text{Gamma}(x, 10, 3) + 0.3 \cdot \text{Beta}(x, 10, 10)$ where x is defined over $(-\infty, \infty)$. I want to draw samples from the above PDF using Metropolis-Hastings Algorithm. The proposal distribution is $N(X(t), s)$, where $X(t)$ is the last accepted sample. Simulate a few samples with a suitable value of s . Illustrate two situations where i) having small value of s is useful, and ii) having large value of s is useful.
7. Provide an algorithm based on importance sampling to carry out the integration of $(\sin(x)/(\log(x)+1)) \cdot x^5 \cdot (1-x)^4$ over the interval (0,1). If your proposal distribution is $N(0,1)$, which samples will have high “importance scores”?

Coding questions:

1. We want to simulate a 3-component Gaussian mixture model with means -3, 0, 8 where each component has equal proportion and equal variance 25. For this, start metropolis-hastings sampling from initial point 10 with a proposal distribution $N(0,9)$. What is the probability that only one Gaussian component gets simulated?

2. Consider the following PDF $f(x) = 0.5 \cdot N(x, -5, 25) + 0.25 \cdot \text{Gamma}(x, 10, 3) + 0.25 \cdot \text{Beta}(x, 10, 10)$ where x is defined over $(-\infty, \infty)$. Consider that Gamma PDF is 0 in $(-\infty, 0)$ and Beta PDF is 0 outside of $(0, 1)$. Verify that this is a valid PDF.
- Choose $x=0$ as the initial point, and use $N(x(t), 10)$ as the proposal distribution for state transition. Use Metropolis-Hastings to draw 1000 samples (ACCEPT+REJECT) from this distribution. How many of these were accepted? Create the histogram of the accepted samples and contrast it with the PDF f .
- Now repeat the experiment using $N(x(t), 100)$ as the proposal distribution for state transition. How does the acceptance rate and histogram of accepted samples change?